Coding & Solution

Date	1 July 2026
Team ID	--------------
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/k-chandravardhanreddy/Rising_waters.git
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	• Flood prediction using Machine Learning • User-friendly web interface • Rainfall data input form • Flood/No Flood prediction result page • Responsive design with animated rainfall background • Trained Random Forest model integration
Pending / Incomplete Features	• User authentication • Historical prediction records • Live rainfall data integration through APIs • Interactive flood visualization maps
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using pip install -r requirements.txt. 3. Run the application using python app.py. 4. Open the local server URL (or deployed Render URL) in a web browser. 5. Enter rainfall values and click **Predict** to view the flood prediction result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

- The project follows a modular Flask application structure for better readability and maintainability.
- A trained Random Forest Machine Learning model is used to predict flood occurrence based on rainfall inputs.
- The application has been successfully deployed and tested.
- Git is used for version control, enabling efficient code management and collaboration.
- The user interface includes a responsive layout with a rainfall animation to enhance user experience